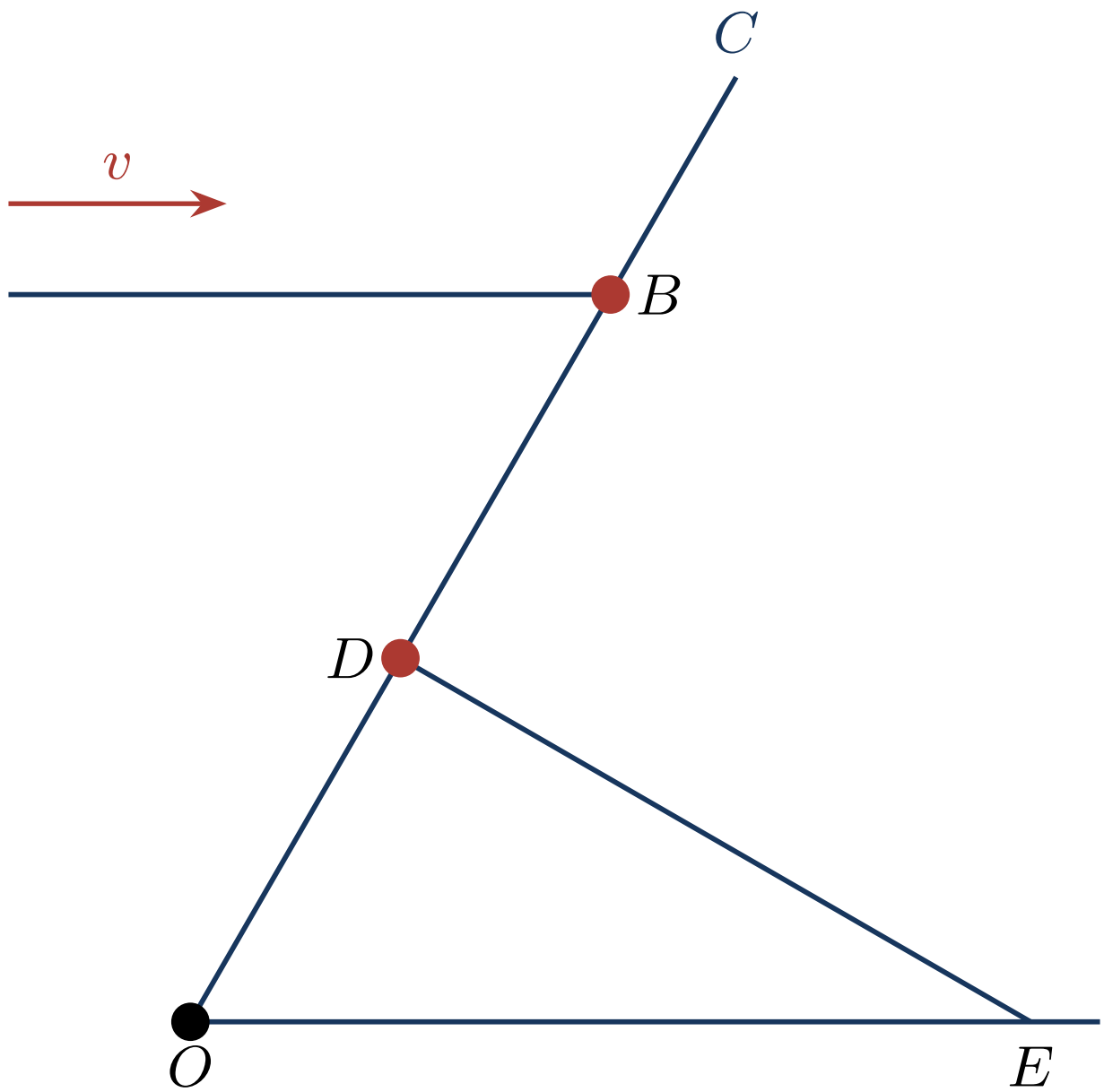
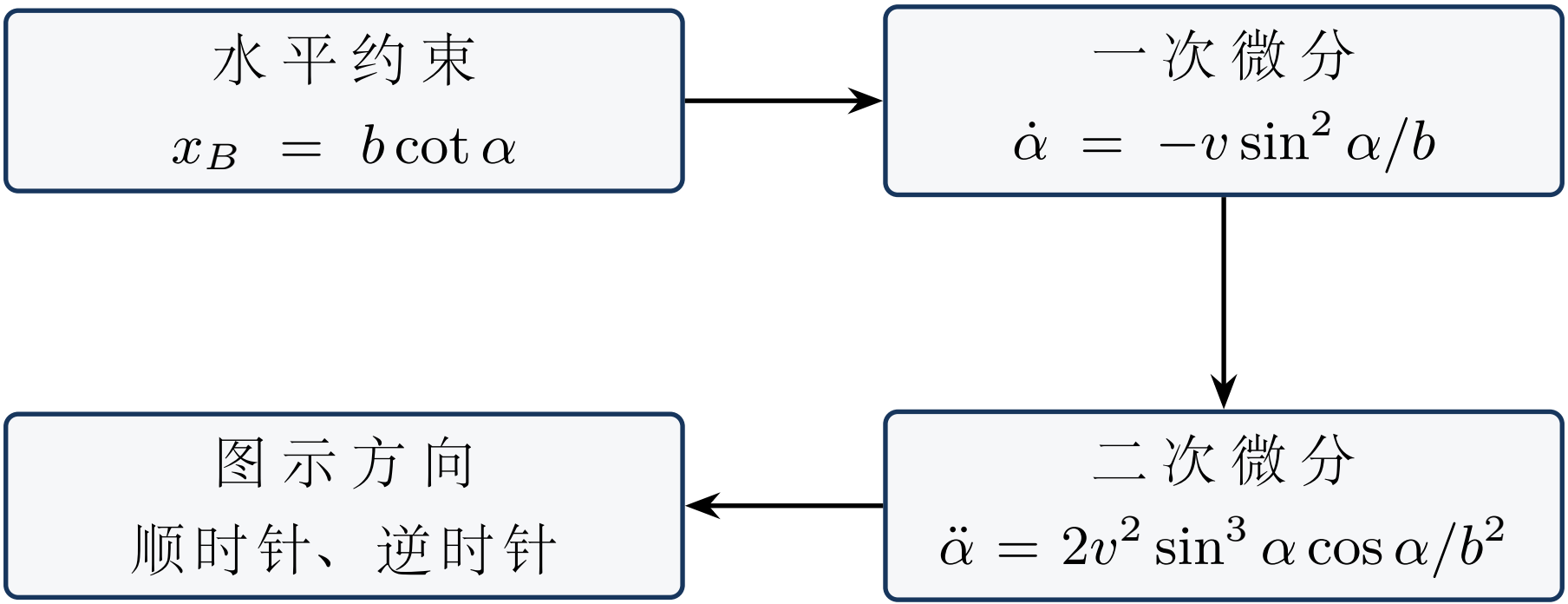
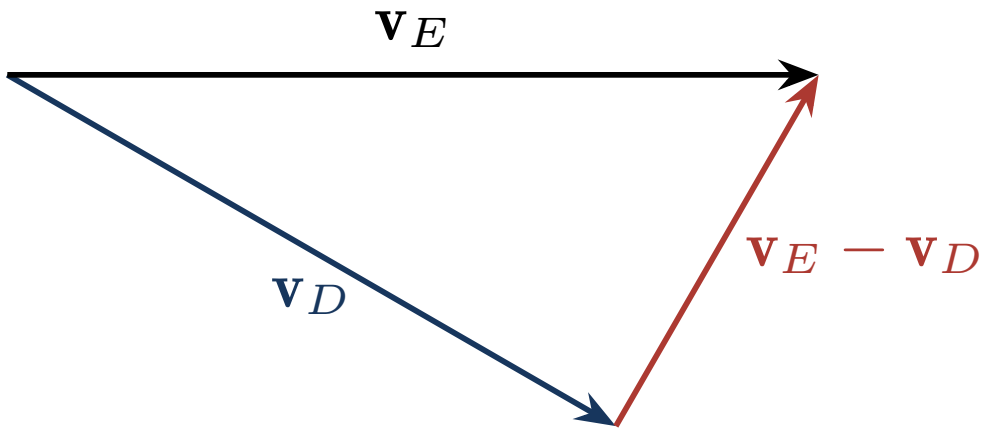


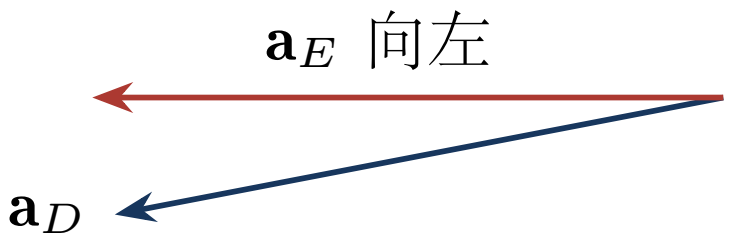
B 沿杆滑动；D 是杆上固定点；$OD = BD$ 仅在图示瞬间



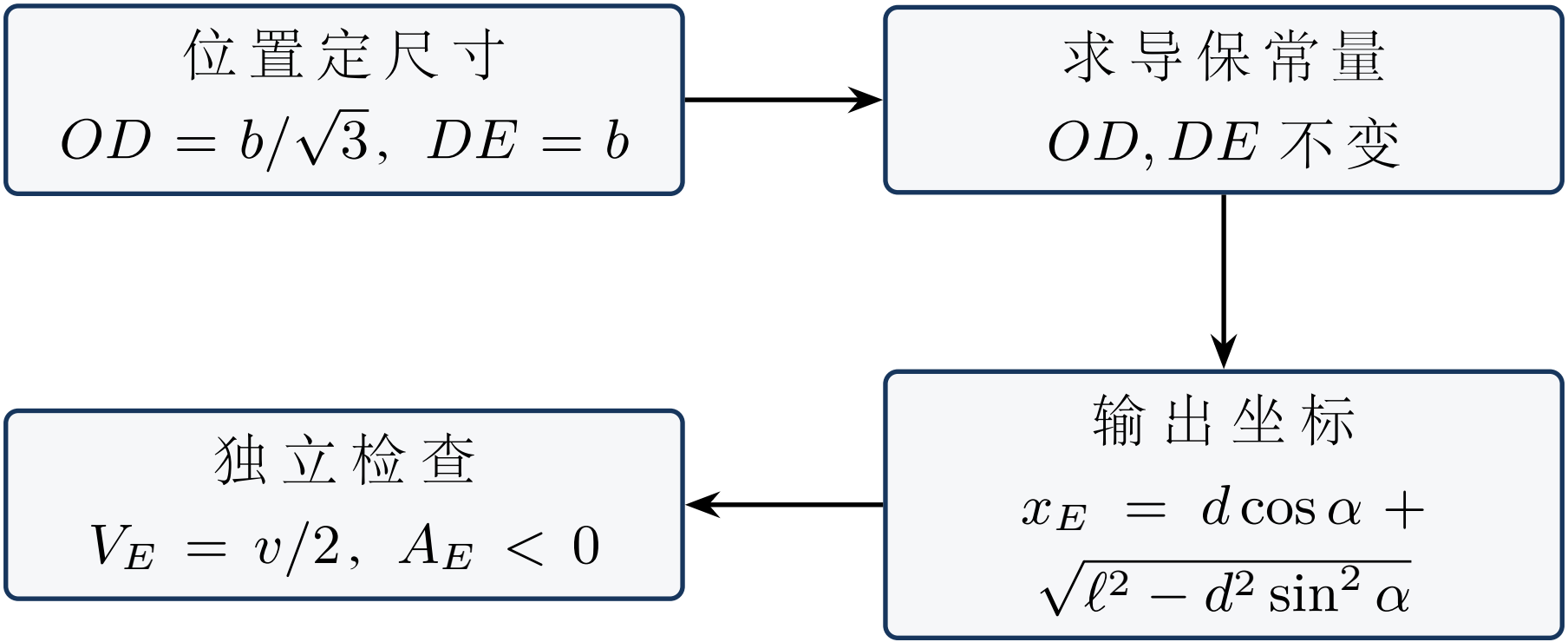
角速度与角加速度可以方向相反



速度分量按 $v/8$ 归一；相对速度垂直于 DE



二阶约束：$(\mathbf{a}_E - \mathbf{a}_D) \cdot \mathbf{r}_{ED} = -|\mathbf{v}_E - \mathbf{v}_D|^2$



瞬时中点不能作为任意时刻的中点约束